

Week 5 Gen AI Internship – Short Report

Project: Personal Expense Tracker

This project demonstrates the Week 5 topic, AI for Programming & Development. The application was developed as a small Python command-line program with AI used as a development assistant rather than a replacement for understanding the code.

1. Objective

Build a small Python application while using AI to generate code, explain code, debug an issue, improve validation and error handling, generate documentation, create tests, and support a final code review.

2. Application Features

Feature	Implementation
Add expense	Amount, category and description are validated before saving.
View expenses	Displays stored records in a readable table.
Summary	Calculates total and category-wise spending.
Persistence	Uses a local JSON file.
Validation	Rejects invalid amounts and invalid/empty text.
Error handling	Handles invalid JSON and file errors.
Testing	10 automated unittest cases.

3. AI-Assisted Development Workflow

- Plan – converted the application idea into functional requirements.
- Code – generated an initial structure and implementation with AI assistance.
- Explain – used AI to explain functions, JSON persistence and exception handling.
- Debug – reviewed a storage failure involving malformed JSON and added safe handling.
- Test – generated and ran unit tests covering normal and invalid cases.
- Improve – manually reviewed readability, validation, error handling and maintainability.

4. Debugging and Verification

A reliability issue was identified in JSON loading: malformed stored data could cause the program to terminate with a JSON decoding error. The final version catches the relevant exceptions and returns an empty list with a warning. A dedicated automated test verifies this behavior.

5. Testing

The test suite contains 10 cases covering valid and invalid amount input, category validation, expense creation, total calculation, category totals, JSON persistence, and corrupted JSON handling. The expected result is that all 10 tests pass.

6. Manual Code Review

- Functions were kept focused and named clearly.
- Input validation was checked for common invalid cases.
- File operations were reviewed for exception handling.
- The application uses only the Python standard library.

- No passwords, API keys, or sensitive credentials are stored.
- AI-generated suggestions were reviewed and verified before inclusion.

7. Deliverables

Python source code, README/documentation, automated test cases, AI prompts, debugging record, code review summary, test results, sample data, and this short PDF report.

8. Conclusion

The project demonstrates how AI can support the software development lifecycle from planning and coding through review, debugging, testing and improvement. The developer remains responsible for understanding and verifying the final code.